

Reconfigurable TE-pass polarizer based on lithium niobate waveguide assisted by $\text{Ge}_2\text{Sb}_2\text{Te}_5$ and silicon nitride

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On-chip polarization management components play a critical role in tackling polarization dependence in the lithium-niobate-on-insulator (LNOI) platform. In this work, we proposed a reconfigurable TE-pass polarizer based on optical phase change material (GST) and the LNOI wafer. The key region is formed by a hybrid GST– Si_3N_4 layer symmetrically deposited atop the centerline of the LNOI waveguide along the propagation direction where the GST is sandwiched in the middle of the Si_3N_4 layer. Whether the polarizer will take effect depends on the phase states of the GST layer and the graphene and aluminum oxide layers are coated atop the GST– Si_3N_4 layer as the microheater to control the conversion of phase states. The proposed device length is $7.5\text{ }\mu\text{m}$ with an insertion loss (IL)=0.22 dB and extinction ratio (ER)=32.8 dB at the wavelength of 1550 nm. Moreover, it also has a high ER (>25 dB) and a low IL (<0.5 dB) in the operating bandwidth of 200 nm. Such a high-performance TE-pass polarizer paves a new way for applications of photonics integrated circuits. © 2023 Optica Publishing Group

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1. INTRODUCTION

Silicon-on-insulator (SOI) as a mature on-chip platform has been widely implemented on photonic integrated circuits (PICs) due to the large relative refractive index difference (~ 2.0) and CMOS compatible process [1–3]. Conversely, its drawbacks are obvious; it is tough to generate a light source and its linear photoelectric effect is absent, which hinders direct modulation of the light field owing to the centrosymmetric crystal structure of silicon [4,5]. In this regard, the thin film lithium-niobate-on-insulator (TFLN) becomes an ideal candidate optical platform that equips various excellent properties such as large electro-optic, thermo-optic, and acoustic-optic coefficients, as well as nonlinear properties, and stable chemical and physical properties that stem from its intrinsic material characteristics [6,7]. In addition, it has a wide transparent window (400 nm–5 μm) and a relatively high refractive index contrast (~ 0.7) for confining light fields [6,8]. The TFLN can be available from the lithium-niobate-on-insulator (LNOI) wafer by utilizing the smart cut fabrication process and the LNOI wafer can conveniently be purchased from commercial companies [9]. By combining these specific effects of the TFLN with the available LNOI wafer, massive high-speed modulation devices have been proposed and the TFLN is becoming a promising

alternative to SOI whether in the classical or quantum domain [8,10]. Unfortunately, lithium niobate is an anisotropy crystal, and its intrinsic birefringence leads to strong polarization dependence. In practical operation systems, the purity of polarization will critically impact the device's performance. Thus, polarization management devices are significant components of this optical integrated platform [11–13].

There are various types of polarization management devices that have been reported, such as polarization beam splitters (PBSs) [14,15], polarization rotators (PRs) [16,17], polarization splitter-rotators (PSRs) [18,19], and polarizers [20–26]. Based on asymmetrical directional couplers structure, PBS types can separate two orthogonal modes in different channels. PRs are able to rotate one polarization mode to another polarization mode [16,17]. These two types are usually combined as PSRs which can rotate the transverse-magnetic (TM) or transverse-electric (TE) portion of the input light to TE or TM and split them into two output ports [18,19]. Polarizers can efficiently dispel the interference mode while retaining the target mode usually in a single waveguide which has a more compact footprint. In this class, the hybrid plasma grating structure is the universal solution that combines the respective advantages of grating and metal in the extinction ability to achieve a high extinction ratio (ER) (e.g., >20 dB) at the cost of large size

and relatively high insertion loss (IL) (e.g., >2 dB) [20–22]. Graphene-based polarizers which employ the electrical absorption principle apply the multilayer graphene structure on a single waveguide to eliminate the undesirable mode [23,24]. Although such devices usually offer a high ER (e.g., >30 dB), they pose a considerable challenge for fabrication. Moreover, other dielectric materials, such as titanium nitride and indium tin oxide (ITO), are also considered great materials to filter the TE or TM mode [25,26] but they usually struggle to balance high performance with simple processes. Generally, all the abovementioned devices are passive that cannot realize reversible operation between dual polarization and single polarization to obtain the desired mode. While such a tunable mechanism can boost the flexibility of an optical communication system by acquiring desired signals programmatically. Also, in the optical computing domain, it can be possible to optionally calculate different data and achieve programmable multiple outputs, improving the computing capacity in disguise.

To achieve the active reconfigurable function in polarizers, phase change materials (PCMs) are identified as great candidates due to a large property contrast between the two-phase states (crystalline and amorphous) that can correspond to two distinct device functions [27–29]. The phase conversions of PCM have high durability and can be carried out within sub-nanosecond by applying optical or electrical pulses [28,29]. Also, PCMs can realize zero static power [28] for photonic switches based on it, and these abovementioned features guarantee the efficient operation of active optical devices. Commonly used PCMs in polarizers are germanium antimony tellurium alloys and vanadium dioxide (VO_2) which can fulfill the requirement of obstructing the interferential state using a larger material extinction coefficient [30,31]. Xie *et al.* reported TM- and TE-pass polarizers based on the $\text{Si}-\text{Sc}_{0.2}\text{Sb}_2\text{Te}_3$ hybrid waveguide that revealed high ER (>40 dB). However, the device loss is larger than 2 dB due to the introduced material absorption loss and requires effective approaches to reduce it [30]. Zhang *et al.* developed tunable TE- and TM-pass polarizers based on VO_2 with ER >20 dB (TM pass) and >30 dB (TE pass) with compact footprints ($<6\ \mu\text{m}$) but at the expense of complex processes and relatively high loss [31]. However, most of the relevant building blocks are built on the SOI wafer, and PCM-based polarizers on the LNOI platform still need to be developed.

In this paper, we propose a reconfigurable TE-pass polarizer based on the LNOI platform assisted by $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (GST) and Si_3N_4 . The mode polarizer section consists of the Si_3N_4 waveguide coating atop the LN waveguide, and the GST block is embedded in the Si_3N_4 waveguide. Based on such a hybrid waveguide structure, the mode polarized effect would take place along the transmission direction. By enabling the operation state (crystalline state) of the GST layer, this application can efficiently filter the TM mode and pass the TE mode with ER = 32.8 dB and IL = 0.22 dB. And the working bandwidth is 200 nm (ER >25 dB) within the total length of $7.5\ \mu\text{m}$. When the GST switches into an amorphous state, such polarization features will disappear and can pass the original input mode. Therefore, this active device can offer an efficient and flexible scheme of polarization management components on the LNOI platform.

2. DEVICE STRUCTURE AND PRINCIPLE

Figure 1(a) demonstrates the stereoscopic diagram of our proposed TE-pass polarizer built from the 700 nm thick X-cut LN wafer. The strip Si_3N_4 layer with an embedded GST rectangular block is deposited atop the LN waveguide at a certain length (L). Figure 1(b) illustrates the enlarged cross section, which contains the width and thickness of LN (W , H) and hybrid GST– Si_3N_4 (W_1 , T) layer including the GST layer (W_2 , T). The sidewall inclination of the LN waveguide is chosen as 83 deg according to the current etching technique [32,33]. The GST layer is sandwiched in the middle of the Si_3N_4 layer, and they are loaded on the upper surface center of the LN waveguide along the transmission direction. A single layer of graphene covers the surface of the work area, and the metal electrodes (titanium) are introduced on both sides of the operation region. They form a microheater to realize the phase state conversion of the material GST and their thicknesses ($H_2 = 100$ nm, $H_3 = 0.35$ nm) are marked in Fig. 1(c), respectively. The single graphene layer is efficient electric heating material that equips high electrical and thermal conductivity with low optical absorption loss due to the Pauli blocking mechanism [34]. The surrounding medium is air. Thus, to prevent the graphene from oxidation, we also add an alumina (Al_2O_3) layer with a thickness of $H_1 (=20$ nm) on it.

The X-cut LNOI wafer is selected which has an ordinary index (no) of 2.211 along the x and y axes, and an extraordinary index (ne) of 2.138 along the z axis, at the central wavelength of 1550 nm. The refractive index of GST is $3.80 + 0.025i$ and $6.63 + 1.089i$ for amorphous and crystalline phase states [28], respectively, which are higher than those of LN and Si_3N_4 (~ 2.0). Owing to the large imaginary part of GST in the crystalline state, the efficient absorption loss for the polarization state can be realized by optimizing structure parameters rationally. The working principle is analyzed as follows: when the GST layer is triggered into the crystalline state, its high refractive index will present a nonuniform distribution in the vertical direction between the Si_3N_4 and LN waveguide chamber in

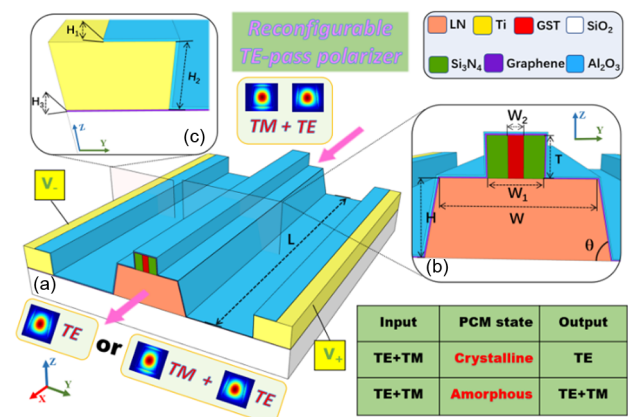


Fig. 1. (a) Schematic of the whole reconfigurable TE-pass polarizer based on LNOI. (b) The cross section of the key polarizer region. (c) Enlarged view of the metal contacting components including the different material layers. For input dual polarization modes (TE and TM), the output can be switched between either single polarization (TE) or dual polarization mode (TE and TM) depending on the phase state of the GST. The material and structural parameters are labeled in the figures.

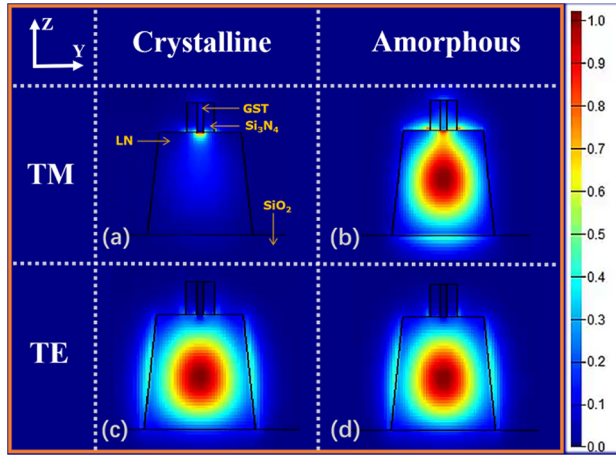


Fig. 2. Light power distribution in the x direction of the proposed device under input TM mode when the GST layer is in (a) the crystalline and (b) amorphous state and input TE mode when the GST layer is in (c) the crystalline and (d) amorphous state, respectively.

which the effective refractive index of the hybrid GST–Si₃N₄ layer will be larger. Thus, considerable power distribution of the TM mode, whose electric field is polarized in the vertical direction, would squeeze into the hybrid GST–Si₃N₄ layer rather than the LN waveguide. Meanwhile, this dense power will be effectively undermined by the GST block within a certain length because of its high material imaginary part, as shown in Fig. 2(a). Conversely, the electric power of the TE mode is concentrated in the LN waveguide instead of the Si₃N₄ waveguide due to its horizontal polarization direction, leading to a weaker interaction between GST and the light field. As a result, the input TM mode will be eliminated by the polarizer region in a relatively short length ($<10\ \mu\text{m}$), but the input TE mode will be transmitted normally with a low loss ($\text{IL} < 0.5\ \text{dB}$), revealed in Fig. 2(c).

On the other hand, when the GST is in the amorphous state that has a lower refractive index, the major light field of the TM mode will be distributed in the LN waveguide. Because of the low extinction coefficient of GST, a small fraction of the light power that conveys in the hybrid GST–Si₃N₄ layer will virtually not lose and continue to be flowed into the LN waveguide after passing through this polarizer region shown in Fig. 2(b). Also, according to the above analysis, the input TE mode will suffer a negligible effect when it passes the polarizer region, as depicted in Fig. 2(d). Therefore, the reconfigurable mechanism for the TE-pass polarizer can realize only by setting the required phase states of GST, where the working region length is less than $10\ \mu\text{m}$. The purpose of using Si₃N₄ material is that its refractive index is close to that of LN, but the thermal conductivity is higher which guarantees the steady transmission of the light field and improves the heating efficiency of the GST phase transition [8,35]. Additionally, the deposition of silicon nitride on lithium niobate films also caters to existing process technologies [36]. The features of this device have been listed in the lower right corner of Fig. 1. In addition, the three-dimensional finite-difference time-domain (3D-FDTD) method [37,38] was used to study and optimize the structural parameters in the following sections.

3. RESULTS AND DISCUSSION

Before we analyze the calculation and optimization of the reconfigurable TE-pass polarizer, the definitions of the key performance indicators ought to be offered. The IL of the TM or TE modes is expressed as [30,31]

$$\text{IL} = -10\log_{10}\left(\frac{P_{\text{out}}}{P_{\text{in}}}\right), \quad (1)$$

where P_{out} and P_{in} stand for the receiving and input power of the TE or TM mode at the device output port and input port, respectively. The ER between TM and TE modes is then defined as [30,31]

$$\text{ER} = \text{IL}_{\text{TM}(\text{cry})} - \text{IL}_{\text{TE}(\text{cry})}, \quad (2)$$

where $\text{IL}_{\text{TM}(\text{cry})}$ and $\text{IL}_{\text{TE}(\text{cry})}$ represent the IL of the output TM mode and TE mode, respectively, when the device is in the operation state. High ER values for TM mode and low IL for the TE mode are our expectations. Meanwhile, low IL for the TE and TM modes is also the desired metric, when GST is in the amorphous state. To simplify the description process, we define the IL_{OT} as three other insertion losses in addition to the $\text{IL}_{\text{TM}(\text{cry})}$, i.e., the IL of the TE mode when the GST is in the crystalline state and the IL of the TM and TE modes in the amorphous state. We defaulted to $1.55\ \mu\text{m}$ as the center calculated wavelength and restricted the workplace in the 3D-FDTD simulation with the grid sizes of $\Delta x = \Delta y = 20\ \text{nm}$, $\Delta z = 10\ \text{nm}$ in x , y , and z directions. The perfectly matched layer boundaries are used at the edges of the computational window to eliminate the nonphysical reflections so that the physics of the device can be accurately captured.

Figure 3 shows the calculated TM and TE mode IL of the proposed device as a function of the length of the GST as well as the Si₃N₄ layer. For the operating state that the GST is triggered into the crystalline state, as marked by the red curve the same as the following figures, the device can implement the function of filtering the TM mode through the TE mode, where the IL of the TM mode exhibits high value ($>25\ \text{dB}$) and low loss for the TE mode ($\text{IL} < 0.5\ \text{dB}$) within the calculation range from 7.1 to $8\ \mu\text{m}$. Such large dimensions greatly broaden the fabrication tolerance. We take the GST in the amorphous state into account and the lower IL_{OT} is locked at $L = 7.5\ \mu\text{m}$, where the TM mode acquired the relatively higher IL in the crystalline state.

Figure 4 shows the effect of the width (W_2) and thickness (T) of the GST layer on the device performance. For W_2 , the GST must be kept within the specified size range due to the current E-beam lithography and etching processes [39,40]. As the W_2 increases, the performance of the TM mode in the crystalline state rises and reaches the peak at $W_2 = 60\ \text{nm}$, and this value also maintains the good performance for IL_{OT} ($<0.5\ \text{dB}$), as indicated in Fig. 4(a). For a fabrication error of $\pm 22\ \text{nm}$, i.e., W_2 varies from 48 to $82\ \text{nm}$, the device performance is acceptable with $\text{IL}_{\text{TM}(\text{cry})} > 19.8\ \text{dB}$ and $\text{IL}_{\text{OT}} < 1\ \text{dB}$. The thickness of the GST and Si₃N₄ layer is set at the optimal value of $T = 200\ \text{nm}$ and when the fabrication error achieves $\pm 35\ \text{nm}$, the performance that $\text{IL}_{\text{TM}(\text{cry})} > 20.0\ \text{dB}$ and $\text{IL}_{\text{OT}} < 0.6\ \text{dB}$ is available. These two parameters should be rigorously controlled during the processing since their fabrication tolerances are relatively tight. At this point, the dimensions

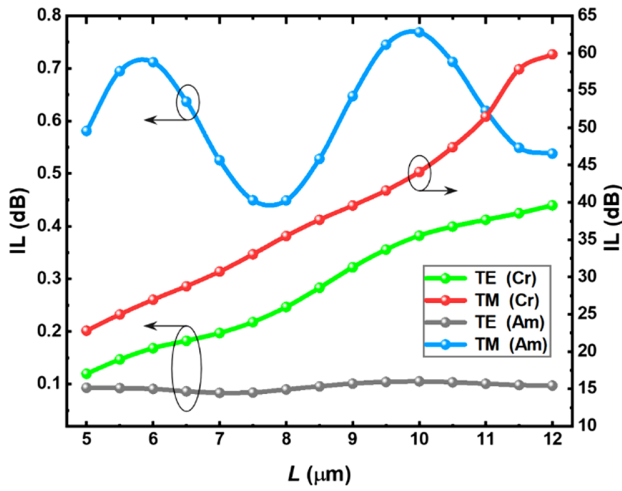


Fig. 3. Calculated IL of the proposed reconfigurable TE-pass polarizer as the function of device length L under input TE and TM modes when the GST layer works in the crystalline (Cr) and amorphous (Am) states, respectively.

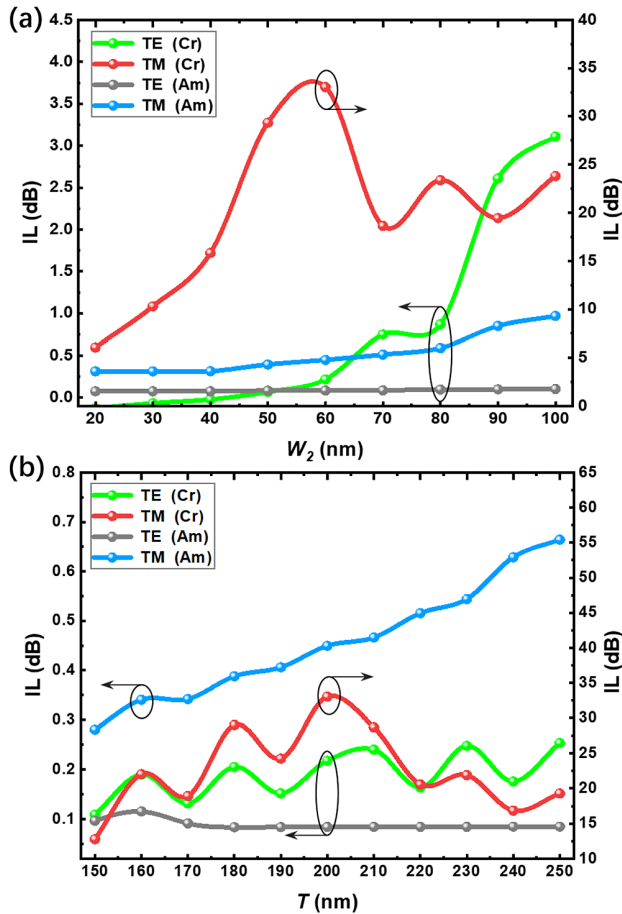


Fig. 4. Device effect (IL) as the functions of the variations of the (a) GST layer width (W_2) and (b) thickness (T), respectively.

of GST have been considered, and high performance can be obtained, where the IL is 33.0 dB (0.22 dB) and 0.45 dB (0.08 dB) for the TM (TE) mode in the crystalline state and amorphous state, respectively.

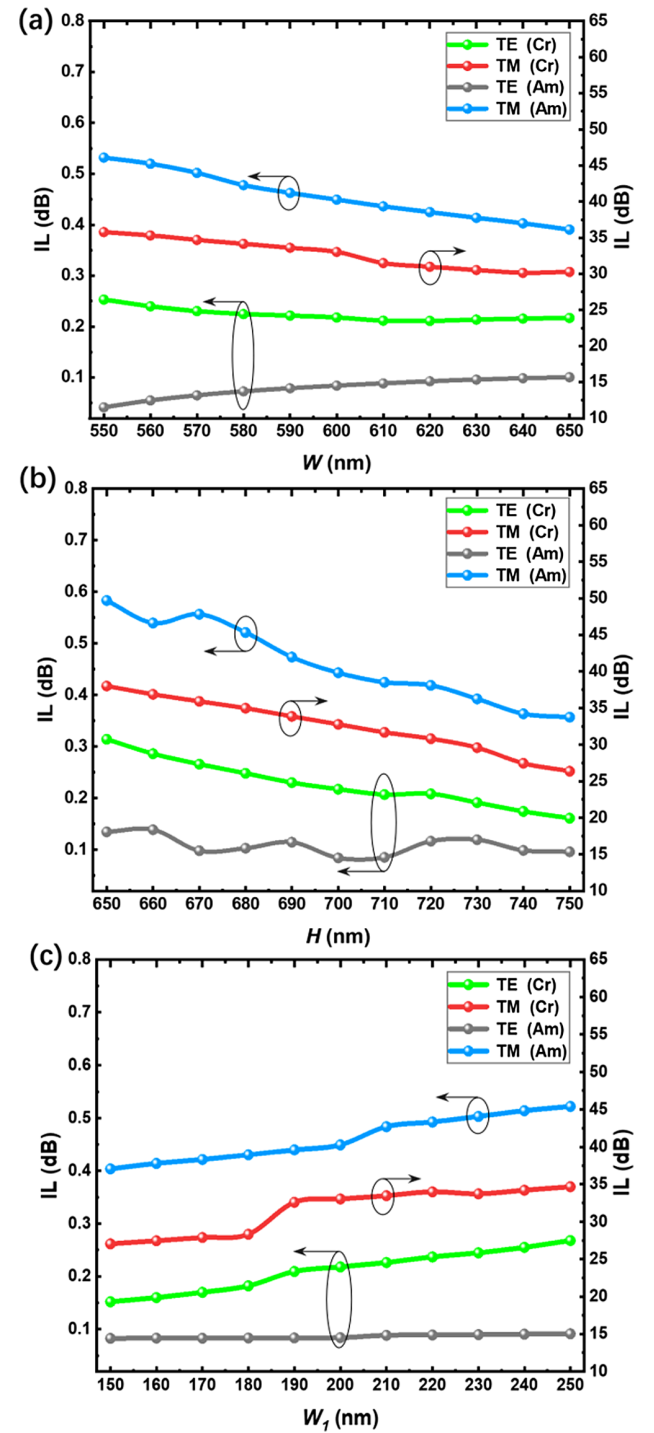


Fig. 5. Device performance (IL) as functions of the (a) width (W), and (b) height (H) of the LN waveguide as well as (c) the width (W_1) of the hybrid GST-Si₃N₄ layer.

Figures 5(a) and 5(b) demonstrate the device performance versus LN waveguide width W and thickness H . The dimensions of the GST layer are fixed as $W_2 = 60$ nm, $T = 200$ nm, and $L = 7.5$ μm , respectively. In order to satisfy the desired criteria ($IL_{\text{TM}(\text{Cr})} > 25$ dB and $IL_{\text{OT}} < 0.5$ dB), the value ranges of W and H are limited to [580 nm, 650 nm] and [690 nm, 750 nm], respectively. Figure 5(c) plots the variation of the hybrid GST-Si₃N₄ layer width W_1 impacts on the IL and

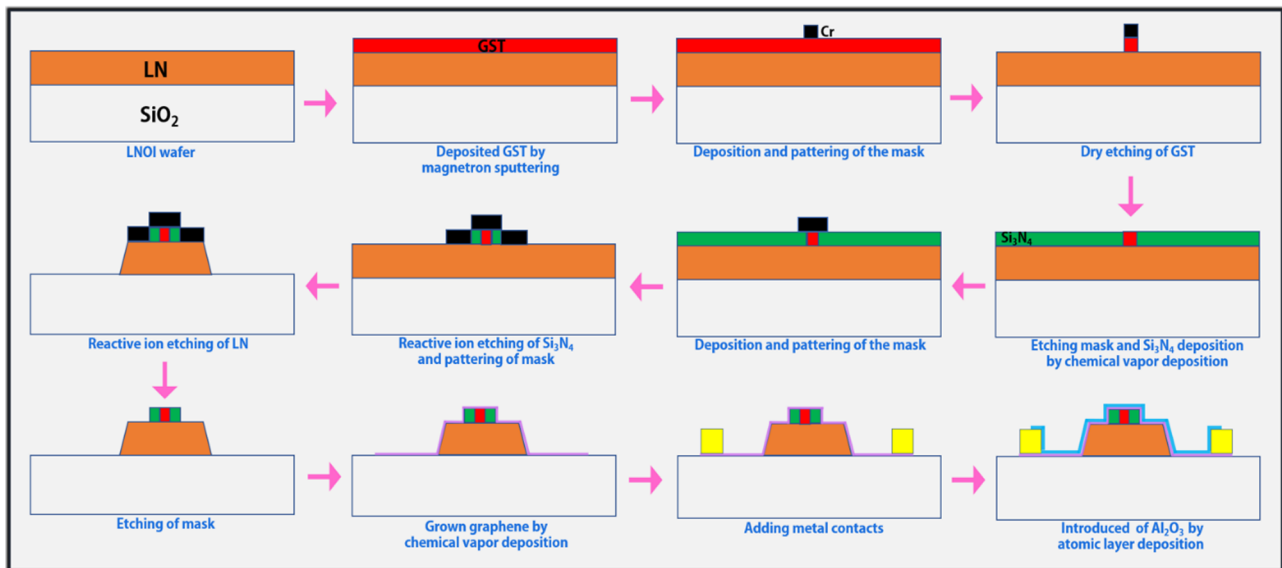


Fig. 6. Process flow diagram of the reconfigurable TE-pass polarizer is directed by arrows, and the annotations for each step of the process are highlighted in blue.

the value corresponding to this parameter has the standard of $IL_{TM(Cry)} > 25$ dB and $IL_{OT} < 0.5$ dB in the interval [150 nm, 220 nm]. The large interval value ranges above mentioned relax the requirements of the fabrication tolerance in the actual manufacturing process. By weighing the performance and compactness of the device, the optimum LN width and thickness are found at $W = 600$ nm and $H = 700$ nm, respectively, and the width of the hybrid GST– Si_3N_4 layer is chosen as $W_1 = 200$ nm.

The major process flow of the proposed reconfigurable TE-pass polarizer is illustrated in Fig. 6 which can be roughly divided into three sections: sputtering and etching out a GST block on the LN wafer, fabricating the deposited Si_3N_4 layer and hybrid structure following a specific size including the strip LN waveguide, and growing graphene as well as the Al_2O_3 layer atop the entire device including the metal electrodes. We start with a commercial standard X-cut LN wafer that consists of a top LN layer thick of 700 nm and a buried oxide layer with a thickness of 2 μ m. First, a GST layer with a width of 60 nm and length of 7.5 μ m is loaded to the top surface of the LN wafer assisted by magnetron sputtering and dry etching [41,42]. Second, by using chemical vapor deposition (CVD), a 200 nm thick Si_3N_4 layer is grown around the GST block [8], and excess Si_3N_4 is etched to form a sandwich structure with a width of 200 nm by utilizing E-beam lithography and reactive ion etching (RIE) processes [36,40], as shown in Fig. 6. Then, a 600 nm wide LN waveguide with an 83 deg sidewall inclination can be fabricated by introducing RIE [10]. Third, a single graphene layer covering the device surface is grown by CVD [35] and the metal contacts are added on both sides of the LN waveguide. Then, to prevent the graphene from oxidating, we introduce a 20 nm layer of Al_2O_3 atop the graphene layer using atomic layer deposition [34]. Following these fabrication steps, the designed reconfigurable TE-pass polarizer can be realized. According to the processing steps, in addition to the parameter tolerances mentioned above, we also need to consider the effect of the

inclination angle, as shown in Fig. 7(a). It can be concluded that when θ varies from 70 to 90 deg, the $IL_{TM(Cry)}$ keeps over 20 dB while the L_{OT} keeps below 0.6 dB, which reflects large fabrication tolerance. In addition, we tested the impact of the center offset (ΔS) of GST on performance during manufacturing, as revealed in Fig. 7(b). For the fabrication error of ± 50 nm, the premium performance can be maintained ($IL_{TM(Cry)} > 20$ dB and $IL_{OT} < 0.5$ dB).

Figure 8 reveals the evolution of the light field along the propagation direction through the proposed reconfigurable TE-pass polarizer calculated by the 3D-FDTD method. From Figs. 8(b) and 8(d), either the TM or TE mode can pass the polarizer region smoothly without any noticeable impairment when the GST is in the amorphous state. Their ILs are 0.45 dB (TM mode) and 0.08 dB (TE mode), respectively, which is mainly caused by the material property of the GST. In Fig. 8(a), when the GST is triggered into the crystalline state, the effect of filtering TM mode will occur due to the strong absorption of the TM mode electric field by the GST after optimizing the structure. Meanwhile, the TE mode will still be output without interference with an IL of 0.22 dB, as shown in Fig. 8(c). From the result, the ER of this device reaches 32.8 dB at the wavelength of 1550 nm, which demonstrates the high matting performance for the input TM mode.

The IL and ER as a function of wavelength spectra for the proposed device are depicted in Fig. 9, which takes each case into account. As a reconfigurable device, the corresponding $IL_{TM(Cry)}$ and IL_{OT} are > 25 dB and < 0.5 dB, respectively, in the calculated wavelength range from 1.45 to 1.65 μ m, which exhibits a small relative wavelength dependence. Thus, the proposed reconfigurable TE-pass polarizer has large allowable operation bandwidths with great performance that cover the main communication bands which surpass many reported comparable devices [14,17,20,23]. To visually highlight the advantages, we listed comparisons of the polarization

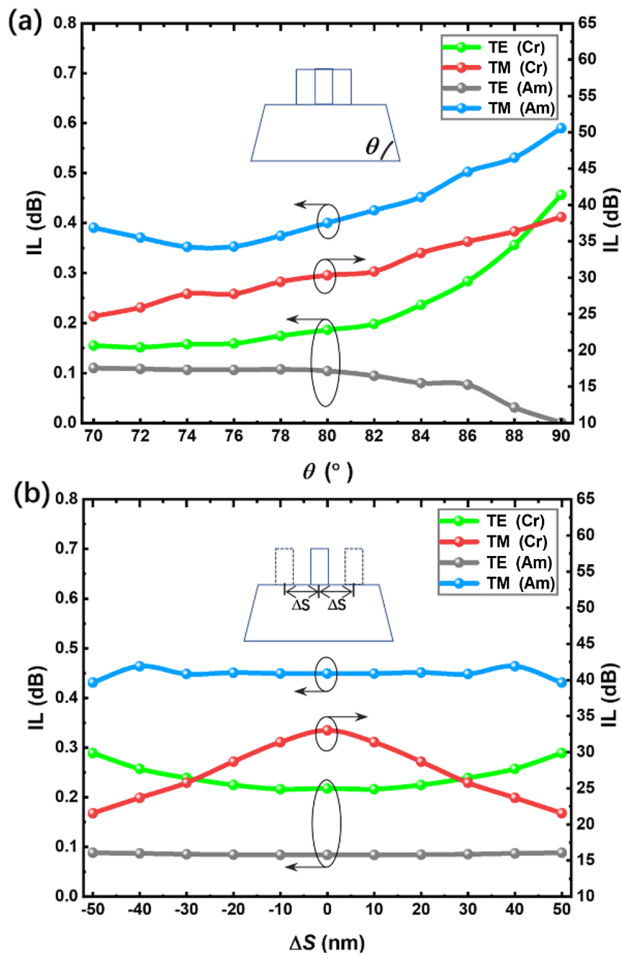


Fig. 7. Device effect (IL) as the functions of the variations of the (a) inclination angle (θ) of LN and (b) the center offset (ΔS) of GST.

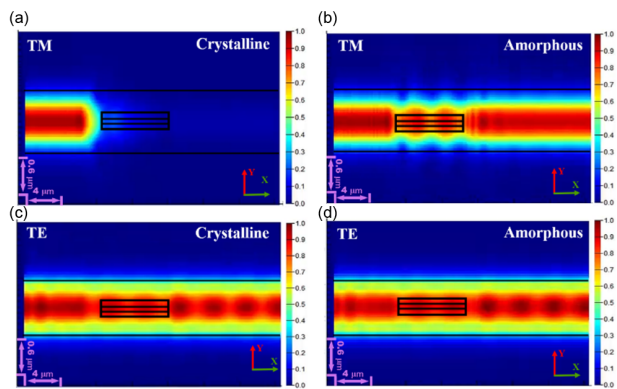


Fig. 8. Optical power distribution of the proposed device under input TM mode, when the GST is in (a) the crystalline state and (b) amorphous state and under input TE mode when the GST works in (c) the crystalline state and (d) amorphous state, respectively.

performance (ER, IL) with other reports, which can be found in the following Table 1.

Next, we analyze the process of the phase change of the GST layer between the crystalline and amorphous states corresponding to the device's reconfigurable feature. The low-power switching can be achieved by heating the PCM intensively using

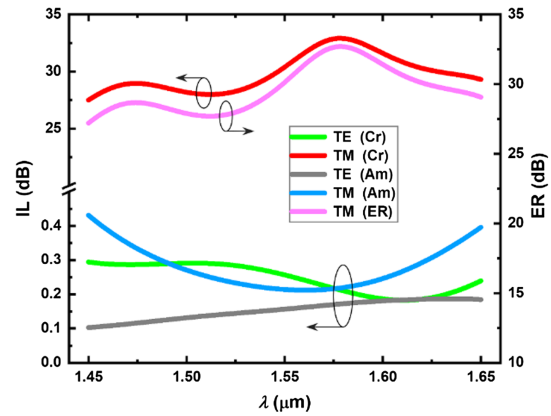


Fig. 9. Wavelength spectra of IL and ER for the proposed reconfigurable TE-pass polarizer under the different cases. The waveguide range is calculated from 1.45 to 1.65 μm .

femtosecond lasers. However, these devices cannot get rid of the external assisted laser, which is not conducive to large-scale integration. For the electrical method, as the microheater and the optical circuit of the device are independent and separate from each other, the devices are more controllable and can enable easier access to complex and multilayered PICs. The polarizers are usually taken effect into the PIC systems, so the electrical method is more suitable. The COMSOL Multiphysics [43] is carried out to calculate the electric-thermal simulation based on the graphene heater, which has proven to exhibit higher heating efficiency than others such as ITO or doped-based [44]. The amorphization processing is the time consumed for raising the temperature of the GST above the melting temperature (877 K) which realizes the phase transition from the crystalline to the amorphous state [45]. A duration pulse is required in this processing to keep the entire 7.5 μm GST layer at a high temperature (>877 K) until ensuring the complete amorphization and then entering the cooling stage. To achieve such an effect, we applied the pulse widths of 100 ns with an 8 ns trailing edge capable of a complete phase transition of the GST layer from the crystalline to the amorphous state driven by 78 mW power. At the moment of 100 ns, the temperature distribution profile of the TE-pass polarizer is shown in Fig. 10(a). It is noted that the melting temperature of GST is lower than those of LN, Si_3N_4 , silica, graphene, and Al_2O_3 so the phase transition process will not damage the device. On the other hand, crystallization that phase conversion from the amorphous to crystalline state can be employed by maintaining the temperature between the crystallization temperature and the melting temperature for a specific time interval [24,25]. To approach the desired requirements, a 1 μs wide pulse with the 1.5 μs trailing edge is attempted which can guarantee long-term crystallization without touching the temperature threshold of amorphous. Figure 10(b) reveals the heated GST layer at the 1 μs moment when the 14 mW power is employed, which further proves the feasibility of this device. In summary, by applying appropriate power pulses, the phase conversion mechanism of GST can be efficiently realized with relative low energy consumption. Also, such a result is not much inferior to the optical tuning method (65 mW power, 50 ns duration for amorphization, and 10 mW power, 2.5 μs duration for crystallization).

Table 1. Comparison of the Proposed Polarizer with Typical Polarizers Reported in the Literature^a

Ref	Function	Length (μm)	ER(dB)	IL (dB)	BW (nm)	RC
[11] (SOI)	TE-pass [E]	30	—	—	60(ER(23–28 dB), IL(2–3 dB))	No
[13] (LNOI)	TE-pass [E]	9	—	—	230 (ER > 20 dB, IL < 2.3 dB)	No
[20] (LNOI)	TM-pass [S]	23	—	—	140 (ER > 20 dB, IL < 2.5 dB)	No
[22] (SOI)	TM-pass [S]	2.5	25	0.088	280 (ER > 10 dB)	No
[24] (SOI)	TM-pass [S]	150	~43	~2.85	200 (ER > 40 dB, IL < 3 dB)	No
[25] (LNOI)	TE-pass [S]	7.5	~22.5	~0.8	125 (ER > 20 dB, IL < 0.95 dB)	No
[31] (SOI)	TE-pass [S]	1.92	~31.9	<0.7	200 (ER > 19 dB)	Yes
	TM-pass [S]	5.95	~20.3	<0.75	200 (IL < 2.2 dB)	
This work (LNOI)	TE-pass [S]	7.5	32.8	<0.5	200(ER > 25 dB, IL < 0.5 dB)	Yes

^aE: experiment; S: simulation; BW: bandwidth; “—”: not mentioned; RC: reconfigurability.

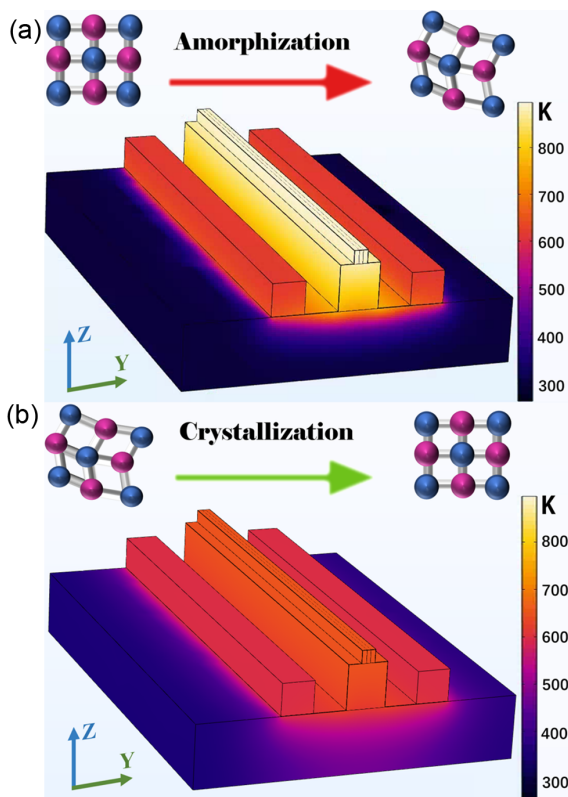


Fig. 10. Electric-thermal simulation for the proposed reconfigurable TE-pass polarizer. The phase state of GST material triggered (a) from the crystalline to the amorphization state by applying a 78 mW power pulse at the moment of 100 ns and (b) from the amorphous to the crystalline state by applying a 14 mW power pulse at the moment of 1 μs , respectively.

4. CONCLUSION

In conclusion, we proposed a reconfigurable TE-pass polarizer based on the LN waveguide, where the reconfigurable scheme is achieved by optical phase change material (GST) embedded into the Si_3N_4 layer. When the GST works at the crystalline

state, a hybrid GST– Si_3N_4 layer deposited on the top center of the LN waveguide can eliminate the TM mode within a length of 7.5 μm . When the GST is in the amorphous state no polarization effect occurs, which corresponds to the reconfigurable function. Note that the TE mode can be transmitted normally in either GST state without interference. From the simulation results, the device performance is $\text{IL}_{\text{TM}(\text{cry})} = 33.0$ dB, $\text{ER} = 32.8$ dB, and $\text{IL}_{\text{OT}} < 0.5$ dB, respectively, at the center wavelength of 1550 nm. And the large working bandwidth was analyzed as well as the device production process. In addition, we also carried out the electric-thermal simulation for the phase change process, and the required power dissipation for GST phase transitions was determined. With these advantages, the proposed device scheme can provide a good sample for optical polarization management components in the integrated photonic circuits at the lithium niobate platform.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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